# Two stochastic epidemic models

We consider two discrete-time, branching-process models for the spread of an infectious disease. Time is measured in days. Let $I_t \geq 0$ denote the number of new infections (incidence) on day $t$, and let $R_t > 0$ be the effective reproduction number on day $t$.

The *serial interval weights* $w_s > 0$ $(s = 1, 2, \ldots)$ satisfy $\sum_{s=1}^{\infty} w_s = 1$ and encode the probability that a secondary infection appears $s$ days after the primary case. Write

$$F_r = \sum_{s=1}^{r} w_s$$

for the cumulative serial interval distribution at lag $r$, so $1 - F_r = \sum_{s=r+1}^{\infty} w_s$ is the probability that transmission occurs after day $r$.

The *dispersion parameter* $k > 0$ controls individual-level variation in infectiousness. Small $k$ corresponds to high heterogeneity (superspreading); as $k \to \infty$ the negative-binomial offspring distribution converges to a Poisson.

**Remark.** Throughout, $\mathrm{NB}(\text{mean} = \mu, \text{disp} = k)$ denotes the negative-binomial distribution with probability mass function

$$\mathbb{P}(X = n) = \binom{n+k-1}{n} \left(\frac{k}{k+\mu}\right)^{k} \left(\frac{\mu}{k+\mu}\right)^{n}, \quad n = 0, 1, 2, \ldots,$$

mean $\mu$, variance $\mu + \mu^2/k$, and probability generating function (pgf) $G(s) = \big(k \,/\, (k + \mu(1-s))\big)^{k}$. Similarly, $\mathrm{Gamma}(\alpha, \beta)$ denotes the Gamma distribution with shape $\alpha$ and rate $\beta$, so density $f(x) = \beta^{\alpha} x^{\alpha-1} e^{-\beta x}/\Gamma(\alpha)$, mean $\alpha/\beta$, and variance $\alpha/\beta^2$.

# SSE model (Superspreading Events)

In the *superspreading-events* (SSE) model, the force of infection on day $t$ is the weighted sum of past incidence,

$$\lambda_t = \sum_{s=1}^{\infty} w_s \, I_{t-s}, \tag{1}$$

and new infections are negative-binomially distributed:

$$I_t \mid \lambda_t \;\sim\; \mathrm{NB}(\text{mean} = R_t \lambda_t,\ \text{disp} = k\lambda_t)\,. \tag{2}$$

The conditional mean of $I_t$ is $R_t \lambda_t$ and its variance is $R_t \lambda_t + R_t^2 \lambda_t / k$, so the NB inflates the Poisson variance by the factor $1 + R_t \lambda_t / k$, capturing superspreading at the population level.

# SSI model (Superspreading Individuals)

In the *superspreading individuals* (SSI) model, heterogeneity enters at the individual level. Each infected individual has a latent infectivity that is Gamma distributed with mean 1 and variance $1/k$.

Each cohort of $I_t$ infected individuals then has an aggregate infectivity

$$Y_t \mid I_t \;\sim\; \mathrm{Gamma}(\text{shape} = k I_t,\ \text{rate} = k)\,, \tag{3}$$

so $\mathbb{E}[Y_t \mid I_t] = I_t$ and $\mathrm{Var}(Y_t \mid I_t) = I_t / k$. New infections arise as a Poisson process driven by past infectivity,

$$I_t \mid (Y_s)_{s<t} \;\sim\; \mathrm{Poisson}\!\left( R_t \sum_{s=1}^{\infty} w_s \, Y_{t-s} \right). \tag{4}$$

**Remark.** Marginalising $Y_t$ from the SSI model recovers a negative-binomial offspring distribution (by the Poisson–Gamma mixture identity; see below), so the two models share the same mean–variance structure at the population level while differing in the *mechanism* generating overdispersion.

## Extinction probability

Henceforth assume the reproduction number is constant, $R_t = R_0$. We study the probability that the epidemic goes extinct, starting from a single infectious case on day 0.

By branching-process theory, the extinction probability $q$ is the smallest non-negative fixed point of the pgf of the offspring distribution. For both models the offspring distribution is $\mathrm{NB}(\text{mean} = R_0, \text{disp} = k)$, whose pgf is $G(s) = (k/(k + R_0(1 - s)))^k$. Setting $q = G(q)$ gives

$$q = \left(1 + \frac{R_0}{k}(1 - q)\right)^{-k}. \tag{5}$$

If $R_0 \leq 1$ the unique solution is $q = 1$ (extinction is certain); if $R_0 > 1$ there is a unique solution $q \in (0, 1)$, so the probability of a major outbreak is $1 - q > 0$.

### Extinction probability following 1 case on day 0, then $r$ days without cases

Suppose we observe a single case on day 0 and no new cases on days $1, \ldots, r$. We compute the updated extinction probability $q_r$ for the residual epidemic.

#### SSE model

Each individual infected on day 0 generates cases at lag $s$ according to an independent $\mathrm{NB}(\text{mean} = R_0 w_s, \text{disp} = k w_s)$ distribution. Observing zero cases on days $1, \ldots, r$ implies that all offspring at lags $s \leq r$ were zero, so only lags $s > r$ remain in play.

A crucial closure property of the NB family is that sums of independent NB random variables with the *same success probability* $p = k_i/(k_i + \mu_i)$ are again NB. Here every lag $s$ yields $p = k w_s/(k w_s + R_0 w_s) = k/(k + R_0)$, which is independent of $s$. Summing over all remaining lags, the index case's remaining offspring distribution is

$$\sum_{s=r+1}^{\infty} \mathrm{NB}(\text{mean} = R_0 w_s,\ \text{disp} = k w_s) = \mathrm{NB}(\text{mean} = R_0(1 - F_r),\ \text{disp} = k(1 - F_r)), \tag{6}$$

with pgf $G_r(s) = \big(1 + (R_0/k)(1 - s)\big)^{-k(1-F_r)} = G(s)^{1-F_r}$. Each such remaining offspring, once born, initiates an independent epidemic governed by the original $\mathrm{NB}(R_0,\, k)$ offspring distribution, and so goes extinct with the standard probability $q$ (the smallest fixed point of $G$). The conditioning event constrains only the index case's offspring at lags $\leq r$ — it does not propagate to later generations — so

$$q_r = G_r(q) = \left(1 + \frac{R_0}{k}(1 - q)\right)^{-k(1-F_r)} = q^{\,1-F_r}. \tag{7}$$

As $r \to \infty$, $F_r \to 1$ and $q_r \to 1$: each additional day without cases raises the extinction probability towards certainty.

#### SSI model

It is more natural here to work with the latent infectivity of the index case. Let $y$ denote the realised infectivity of the single case on day 0, with prior

$$y \sim \mathrm{Gamma}(k,\, k), \qquad \mathbb{E}[y] = 1. \tag{8}$$

Conditional on $y$, infections at each lag $s$ are independent with

$$I_s \mid y \ \sim\ \mathrm{Poisson}(R_0\, w_s\, y). \tag{9}$$

Since $y$ is unobserved, we update its distribution given the zero observations before computing the offspring distribution.

**Step 1: posterior update given no cases on days $1, \dots, r$.** By Bayes' theorem, with the $I_s$ conditionally independent Poisson given $y$,

$$
\begin{aligned}
f(y \mid I_1 = \cdots = I_r = 0) &\propto f(y)\, \mathbb{P}(I_1 = \cdots = I_r = 0 \mid y) \\
&= \frac{k^k}{\Gamma(k)}\, y^{k-1} e^{-ky} \prod_{s=1}^{r} e^{-R_0 w_s y} \\
&= \frac{k^k}{\Gamma(k)}\, y^{k-1} \exp\left(-y\left[k + R_0 \sum_{s=1}^{r} w_s\right]\right) \\
&= \frac{k^k}{\Gamma(k)}\, y^{k-1}\, e^{-(k+R_0 F_r)\, y}. \qquad (10)
\end{aligned}
$$

This is proportional to a Gamma$(k,\ k + R_0 F_r)$ density, so

$$
y \mid (I_1 = \cdots = I_r = 0) \ \sim\ \text{Gamma}(k,\ k + R_0 F_r). \qquad (11)
$$

The posterior rate $k + R_0 F_r > k$ reflects the fact that observing no early transmission is evidence that $y$ is small: the index case is likely a low-infectivity individual.

**Step 2: marginal offspring distribution for lags $s > r$.** Given $y$, the total remaining offspring is

$$
I_{\text{rem}} \mid y \ =\ \sum_{s=r+1}^{\infty} I_s \ \sim\ \text{Poisson}(R_0(1 - F_r)\, y)\,. \qquad (12)
$$

To marginalise over the posterior, set $\lambda = R_0(1 - F_r)\, y$. Since $y \sim \text{Gamma}(k,\ k + R_0 F_r)$, the scaling property ($cX \sim \text{Gamma}(\alpha, \beta/c)$ for $X \sim \text{Gamma}(\alpha, \beta)$) gives

$$
\lambda \ \sim\ \text{Gamma}\left(k,\ \frac{k + R_0 F_r}{R_0(1 - F_r)}\right). \qquad (13)
$$

The *Poisson–Gamma mixture identity* states that if $X \mid \lambda \sim \text{Poisson}(\lambda)$ and $\lambda \sim \text{Gamma}(\alpha, \beta)$, then marginally $X \sim \text{NB}(\text{size} = \alpha, \text{mean} = \alpha/\beta)$. Applying this,

$$
I_{\text{rem}} \ \sim\ \text{NB}\left(\text{size} = k,\ \text{mean} = k \cdot \frac{R_0(1 - F_r)}{k + R_0 F_r}\right). \qquad (14)
$$

Compared to the unconditional mean $R_0(1 - F_r)$, the effective mean is reduced by the factor

$$
\frac{k}{k + R_0 F_r} \ <\ 1, \qquad (15)
$$

which arises entirely from the Bayesian update: the zero streak tells us the index case probably has low infectivity.

**Step 3: extinction probability.** Let $G_r$ denote the pgf of $I_{\text{rem}}$. Each remaining offspring, once born, initiates an independent epidemic with a fresh latent infectivity drawn from the prior Gamma$(k, k)$, so it is governed by the original NB$(R_0,\ k)$ offspring distribution and goes extinct with the standard probability $q$. The conditioning constrains only the index case's own offspring counts, not those of later generations, hence

$$
q_r = G_r(q) = \left(1 + \frac{R_0(1 - F_r)}{k + R_0 F_r}(1 - q)\right)^{-k}. \qquad (16)
$$